IT3708 - Bio-Inspired Artificial Intelligence

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THIS IS A VERSION FOCUSED ON THE TEST PART OF THE LECTURE ONLY, RELEASED ON 12.5.2025 AS PART OF EXAM PREPARATIONS IN 2025.

Question 1

☐ 1. **Multiple Choice: Among the following options, which op...**

Points: 1

Question	Among the following options, which option is not considered a "pillar of evolution"?
Answer	Population Diversity Heredity Constraints Selection

Question 2

☐ **2. Multiple Choice: Among the following operations, which...** Points: 1

Question	Among the following operations, which is not part of an evolutionary algorithm's generation cycle?
Answer	Mutation Evaluation Initialization Crossover Selection Reproduction

Question 3

☐ 3. **Multiple Choice: In which type of genetic algorithm (G...**

Points: 1

Question	In which type of genetic algorithm (GA) is one individual generated per generation, replacing one individual in the population?
Answer	Crowding GA
	Generational GA
	Steady-state GA
	Messy GA
	Elitist GA

Question 4

☐ 4. **Multiple Choice: What is a potential unfortunate conse...**

Points: 1

Question	What is a potential unfortunate consequence of too high selection pressure in an evolutionary algorithm that is performing optimization?
Answer	The population converges to individuals at or close to a local optimum that is not a global optimum.
	The fitness of the population does not increase much.
	All individuals in the population converge to be exactly alike, at a local optimum that is not a global optimum.
	All individuals in the population converge to be exactly alike, at a global optimum.
	The population converges to individuals at or close to a global optimum.

Question 5

☐ **5. Multiple Choice: Which of the following is not conside...** Points: 1

Question	Which of the following is not considered a space in which to preserve diversity in an evolutionary algorithm?
Answer	Algorithm space Probability space Genotype space Phenotype space

Question 6

☐ **6. Multiple Choice: Which of the following is not conside...** Points: 1

Question	Which of the following is not considered a diversity-preserving genetic algorithm?
Answer	Deterministic crowding genetic algorithm Fitness sharing genetic algorithm Probabilistic crowding genetic algorithm Simple genetic algorithm Island model parallel genetic algorithm

Question 7

☐ 7. **Multiple Choice: In the 8-queens problem, as discussed...**

Points: 3

Question

In the 8-queens problem, as discussed in class, there are three types of constraints: horizontal, vertical and cross. To solve the 8-queens problem, we consider a genetic algorithm using a permutation representation. In this permutation representation, the i-th position in the chromosome represents the i-th column on the chessboard and the value there represents a row on the chessboard. Consider, for example, the chromosome (1, 3, 5, 2, 6, 4, 7, 8). Here, the "2" in the 4th position of the chromosome means there is a queen in the 2nd row for the 4th column of the chessboard.

Mutation and crossover work as follows, as discussed in class: Mutation takes, for example, (1, 3, 5, 2, 6, 4, 7, 8) and creates (3, 1, 5, 2, 6, 4, 7, 8). Crossover takes, for example, parents (1, 3, 5, 2, 6, 4, 7, 8) and (8, 7, 6, 5, 4, 3, 2, 1), and creates children (1, 3, 5, 4, 2, 8, 7, 6) and (8, 7, 6, 2, 4, 1, 3, 5). The fitness function takes care of remaining constraint violations, if any, by counting the number of constraints violated. Fitness of an individual is inversely proportional to the number of constraints violated.

How can we describe the constraint handling of the genetic algorithm in this 8-queens problem?

Answer

All constraints are handled by the fitness function

The cross and vertical constraints are handled by the fitness function, while the horizontal constraints are handled by other GA operators

The vertical constraints are handled by the fitness function, while the cross and horizontal constraints are handled by other GA operators

The cross and horizontal constraints are handled by the fitness function, while the vertical constraints are handled by other GA operators

The horizontal constraints are handled by the fitness function, while the cross and vertical constraints are handled by other GA operators

The cross constraints are handled by the fitness function, while the horizontal and vertical constraints are handled by other GA operators

Question 8

☐ 8. **Multiple Choice: Suppose that we have a problem in whi...** Points: 1

Question	Suppose that we have a problem in which a genetic algorithm with a permutation representation is being used. For example, the problem may be the 8-queens problem or the traveling salesman problem (TSP). Which of the following crossover operators would perform poorly for this problem?
Answer	Edge recombination crossover
	Partially mapped crossover
	Order 1 crossover
	One-point crossover

Question 9

☐ 9. **Multiple Choice: Which of the following types of Marko...**

Points: 1

Question	Which of the following types of Markov chains is most commonly used to analyze hill climbing and genetic algorithms that operate in a search space of bitstrings?
Answer	Discrete state space, continuous time Discrete state space, discrete time Continuous state space, continuous time Continuous state space, discrete time

Question 10

☐ 10. **Multiple Answer: Which of the following options are tr...** Points: 1

Question	Which of the following options are true for a homogenous Markov chain? (This is a multiple answer question.)
Answer	The 1-step transition probability matrix can never change
	The 1-step transition probability matrix can change
	All entries in the 1-step transition probability matrix are greater than zero
	All entries in the 1-step transition probability matrix are greater than or equal to zero

Question 11

☐ **11. Multiple Choice: Consider a population of size s, cons...**

Points: 2

Question	Consider a population of size s, consisting of individuals that are bitstrings of length n. The population is initialized uniformly at random. Suppose that n = 5 and s = 4. What is the probability that the i-th bit of each individual in the population is the same for a given i?
Answer	Too little information is provided in the question to give an answer. 0,5 0,25 0,125 0,0625 0,03125

Question 12

Points: 3

12. Multiple Choice: Supppose that we have a genetic algor...

Question	Supppose that we have a genetic algorithm with a population of 10 individuals (x_1, ..., x_i, ..., x_10). Suppose that the fitness f of the i-th individual is f(x_i) = i, and that roulette wheel selection is used to select individuals for crossover. Selection is done with replacement. What is, approximately, the probablility that x_10 is selected to cross over with itself when two parents are selected from the popuation of 10?
Answer	0%
	0.9%
	1.7%
	2.1%
	3.3%
	9.5%
	16.1%
	18.2%

Question 13

☐ **13. Multiple Answer: Suppose that we are using a permutati...** Points: 2

Question	Suppose that we are using a permutation representation and have the following individual: (1, 2, 3, 4, 5, 6, 7, 8, 9). Which of the following individuals, if any, could result from applying swap mutation? (This is a multiple answer question.)
Answer	None. Swap mutation does not make sense for the given individual.
	(1, 5, 3, 4, 2, 6, 7, 8, 9)
	(1, 5, 3, 4, 2, 7, 6, 8, 9)
	(1, 2, 3, 4, 5, 7, 6, 8, 9)
	(1, 2, 3, 4, 5, 5, 7, 8, 9)
	(9, 2, 3, 4, 5, 6, 7, 1, 8)

Question 14

☐ **14. Multiple Answer: Suppose that we are using a permutati...** Points: 2

Question	Suppose that we are using a permutation representation and have the following individual: (1, 2, 3, 4, 5, 6, 7, 8, 9). Which of the following individuals, if any, could result from applying scramble mutation? (This is a multiple answer question.)
Answer	(1, 2, 3, 5, 2, 6, 7, 8, 8)
	(9, 8, 7, 6, 5, 4, 3, 2, 1)
	(1, 3, 5, 4, 2, 6, 7, 8, 9)
	(1, 3, 5, 5, 3, 6, 7, 8, 9)
	(2, 1, 3, 4, 6, 5, 7, 8 , 9)
	(1, 2, 3, 4, 5, 6, 9, 8, 7)

Question 15

☐ **15. Multiple Answer: We are interested in problems that ma...**

Points: 2

Question	We are interested in problems that may or may not have fitness functions and may or may not have constraints. In such problems, in which of the following settings is it natural to use an evolutionary algorithm? (This is a multiple answer question.)
Answer	In no such setting.
	In the constrained optimization setting, where we handle both constraints and a fitness function during the operation of the EA.
	In the constraint satisfaction setting, where we handle constraints during the operation of the EA but there is no fitness function.
	In the free optimization setting, where we only handle a fitness function during the EA's opration. Constraints, if any, are taken care of and integrated into the fitness function in a preprocessing step.
	In the free optimization setting, where we only handle a fitness function during the EA's opration. Constraints, if any, are disregarded in a preprocessing step.

Question 16

☐ 16. **Multiple Answer: We consider in this question the pare...** Points: 2

Question	We consider in this question the parent selection step in evolutionary algorithms. Which of the following selection methods rely on knowledge of the entire population prior to selection? An example of " knowledge of the entire population" would be knowing the fitness of all individuals in the whole population, even if the population is distributed across multiple computers or multiple computational processes. (This is a multiple answer question.)
Answer	Selection in the island model Fitness proportional selection Tournament selection Fitness sharing Ranking selection

Question 17

☐ **17. Multiple Answer: A problem instance can be solved by s...** Points: 2

Question	A problem instance can be solved by some algorithm (with no restrictions on running time) and any solution can be verified by another algorithm in polynomial time. Which complexity class or classes does this describe? (This is a multiple answer question.)
Answer	No complexity class is described in this way.
	This complexity class exists, but it is not among the ones listed below.
	Class P
	Class NP
	Class NP-complete
	Class NP-hard

Question 18

☐ **18. Multiple Answer: Consider a GA with a population of th...**

Points: 3

Question	Consider a GA with a population of these bistrings: 01101, 11000, 01000, and 10011. Further suppose that we have the following values for the fitness function f: • f(01101) = 15 • f(11000) = 40 • f(01000) = 8 • f(10011) = 37 We assume that the population size is constant. What are the expected number of copies of individuals 01101 and 11000 after parent selection? (This is a multiple answer question.)
Answer	0.08 0.15 0.32 0.37 0.4 0.6 1.48 1.6